BTS-IV(R/S)-06-24-2780 Reg. No.

B

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

IT 19-204-0402 DATA COMMUNICATION AND NETWORKING

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Identify the requirements for a high order communication systems.
CO2: Learn with a solid foundation in fundamentals required to have a better understanding of the Internet.
CO3: Understand the working of network protocols and standards.
CO4: Explain the functionalities and protocols of various layers in ISO/OSI Network model.
CO5: Use a suitable transport/application layer protocol based on application requirements.
CO6: Suggest an appropriate access control, congestion control and congestion avoidance technique for a given traffic scenario.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	We need to send 265 kbps over a noiseless channel with a band width of 20 kHz. How many signal levels do we need?	3	L1	1	1,2,12
	(b)	What does the amplitude of a signal measure? What does the frequency of a signal measure? What does the phase of a signal measure?	3	L2	2	1,2,3,12
	(c)	What is the Hamming distance? What is the minimum Hamming distance?	3	L2	3	1,2,12
	(d)	Compare and contrast byte-stuffing and bit-stuffing. Which technique is used in byte-oriented protocols? Which technique is used in bit-oriented protocols?	3	L1	3	1,2,3,4,12
	(e)	What is the difference between the delivery of a frame in the data link layer and the delivery of a packet in the network layer?	3	l3	4	1,2,3,4,12
	(f)	What are the functions of a RIP message?	3	L2	4	1,2,3,12
	(g)	Do port addresses need to be unique? Why or why not? Why are port addresses shorter than IP addresses?	3	L2	5,6	1,2,3,4,12
	(h)	What are the three domains of the domain name space?	3	L1	5,6	1,2,3,4,12

PART B

		(4 × 12 = 48)				
II.		Briefly describe the layers in the ISO-OSI reference model.	12	L2	2	1,2,3,4,12
		OR				
III.	(a)	Explain data rate limits of noiseless channel and noisy channel in data communication.	6	L2	2	1,2,3,4,12
	(b)	Explain the classification of transmission medias used in physical layer.	6	L2	2	1,2,3,4,12

(P.T.O.)

			Marks	BL	CO	PO
IV.		What are the different types of error detecting codes? Explain each with an example.	12	L3	3	1,2,3,4,12
		OR				
V.		Briefly explain the protocols for noisy channels in data link layer.	12	L3	3	1,2,3,4,12
VI.		What is address mapping? Explain dynamic mapping logical to physical address.	12	L2	4	1,2,3, 4,12
		OR				
VII.		What do you mean by distance vector routing? Explain the operation, characteristics and key properties of distance vector routing algorithm.	12	L2	4	1,2,3,4,12
VIII.		Explain the protocols using in transport layer.	12	L2	5,6	1,2,3
		OR				
IX.	(a)	How to map the domain names to the IP addresses using DNS protocol.	6	L2	5,6	1,2,3,4,12
	(b)	Compare the TCP header and the UDP header. List the fields in the TCP header that are missing from UDP header. Give the reason for their absence.	6	L3	5,6	1,2,3,4,12

Bloom's Taxonomy Levels
L1- 31%, L2- 50%, L3-19%.

***